eventbrite

IS6420: Database Theory & Design

Group 2: Danica Sheffield, Jenny Ngo, Sina Odejinmi, Stefan Spasovski

Spring 2026

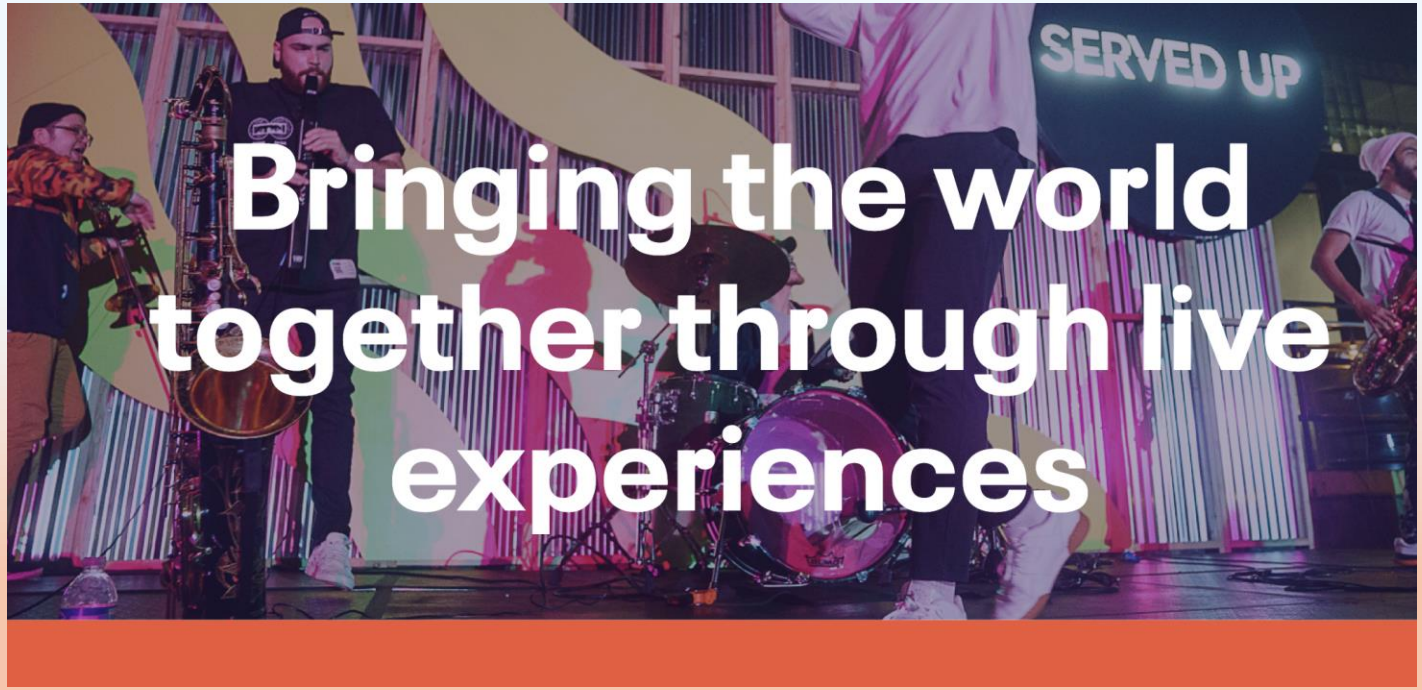
Group 2 Presentation.mp4

Agenda

- Overview of Eventbrite
- Initial Requirements
- Conceptual Model
- Logical Model
- Database Implementation
- Database Demonstration

Eventbrite Overview

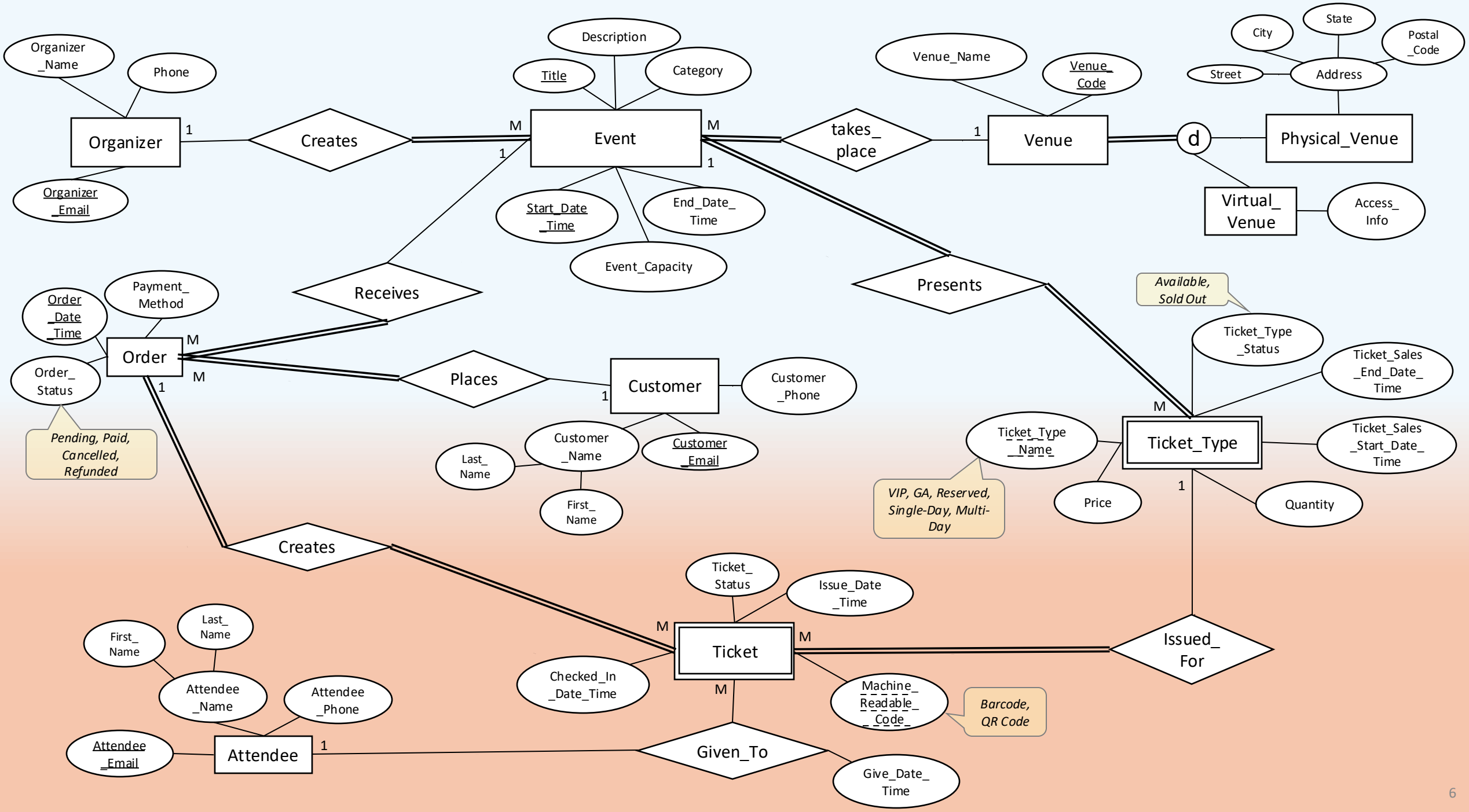
- Create
- Promote
- Manage
- Sell



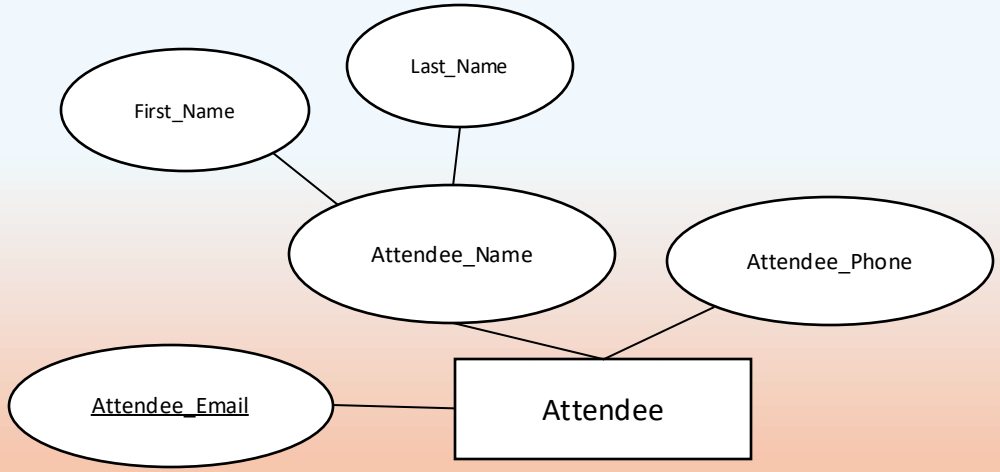
Initial Requirements

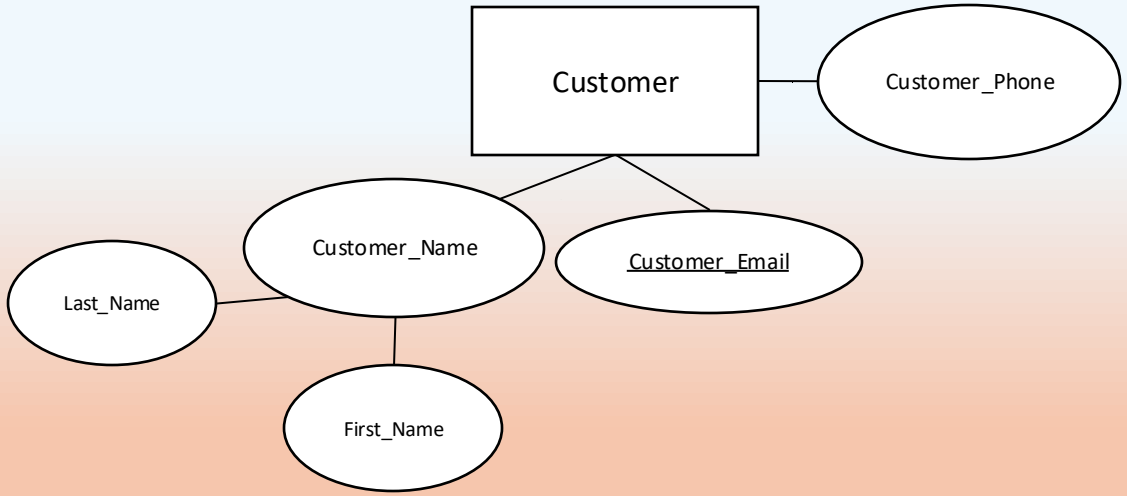
- Organizer
  - Create events
  - Sell tickets
  - Manage events
- Customer
  - Browse for Events
  - Buy Tickets
  - Check-In

Conceptual Model

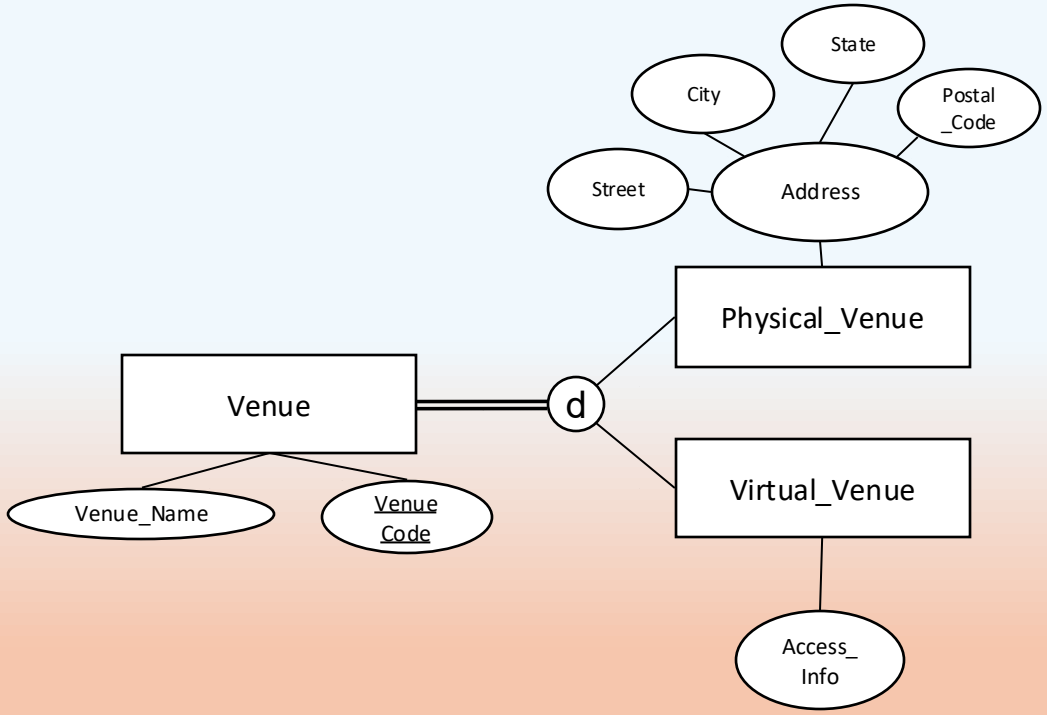

Organizer
Organizer_Name
Phone
Organizer_Email
Creates
Event
Title
Description
Category
Start_Date_Time
End_Date_Time
Event_Capacity
takes_place
Venue
Venue_Name
Venue_Code
d
Physical_Venue
Address
Street
City
State
Postal_Code
Virtual_Venue
Access_Info
Receives
Order
Order_Date_Time
Payment_Method
Order_Status
Pending, Paid, Cancelled, Refunded
Places
Customer
Customer_Phone
Customer_Name
Customer_Email
Last_Name
First_Name
Presents
Ticket_Type
Ticket_Type_Name
VIP, GA, Reserved, Single-Day, Multi-Day
Ticket_Type_Status
Available, Sold Out
Ticket_Sales_End_Date_Time
Ticket_Sales_Start_Date_Time
Price
Quantity
Creates
Ticket
Ticket_Status
Issue_Date_Time
Checked_In_Date_Time
Machine_Readable_Code
Barcode, QR Code
Issued_For
Given_To
Give_Date_Time
Attendee
Attendee_Email
Attendee_Name
Attendee_Phone
First_Name
Last_Name
1
M

Why Attendee ≠ Customer

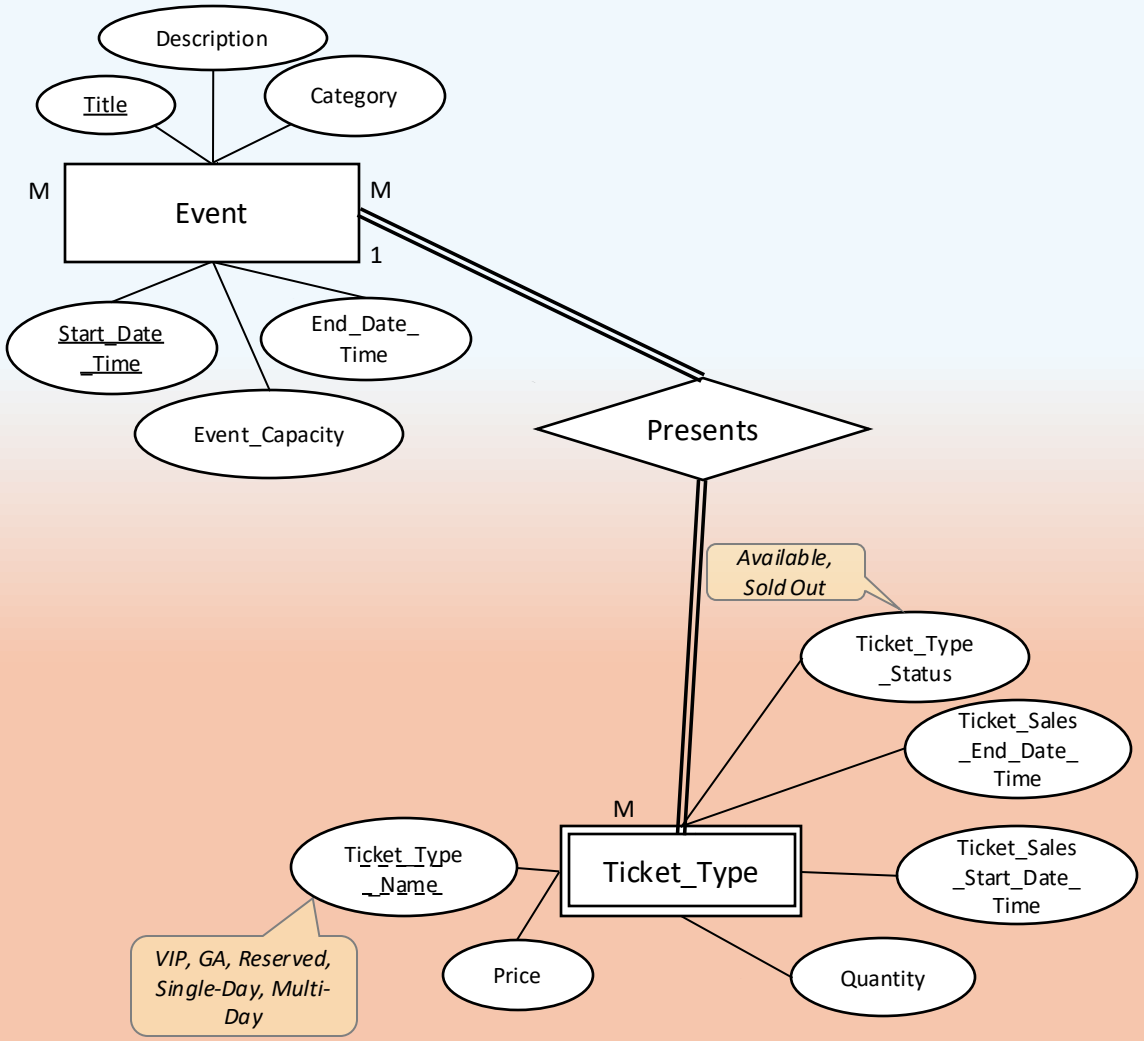


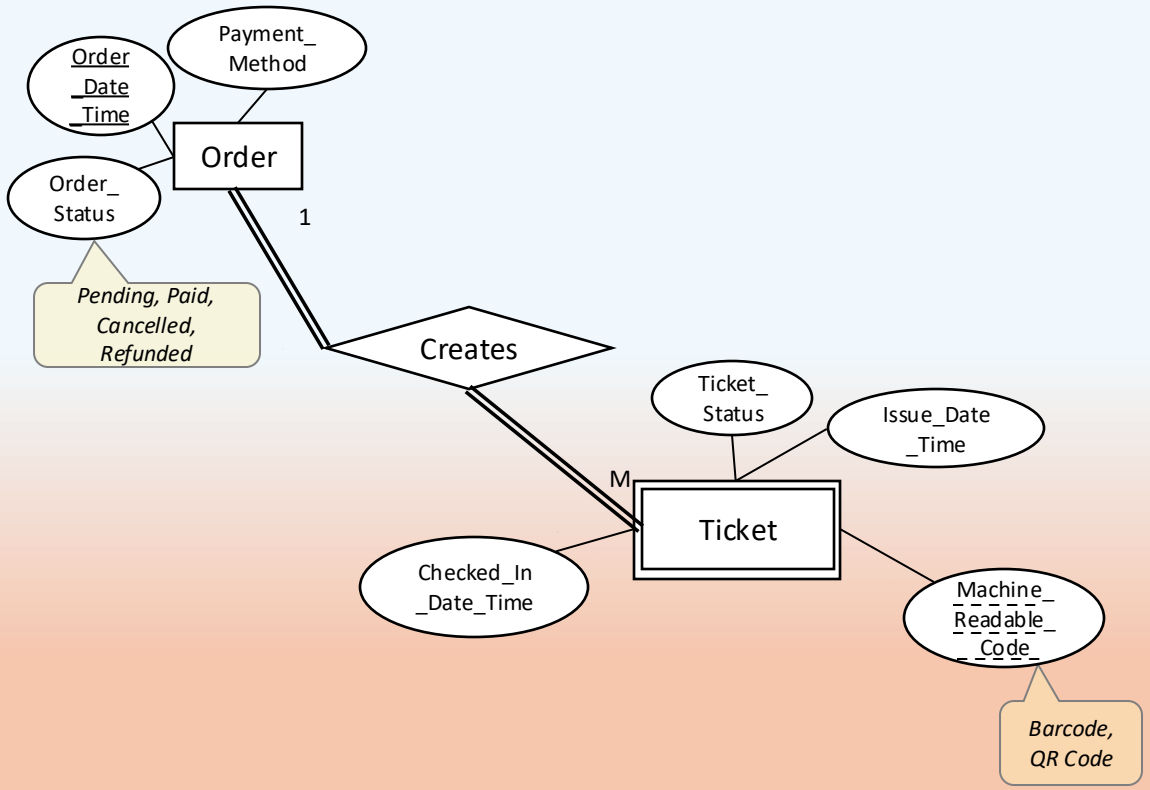


Venue Supertype/Subtype

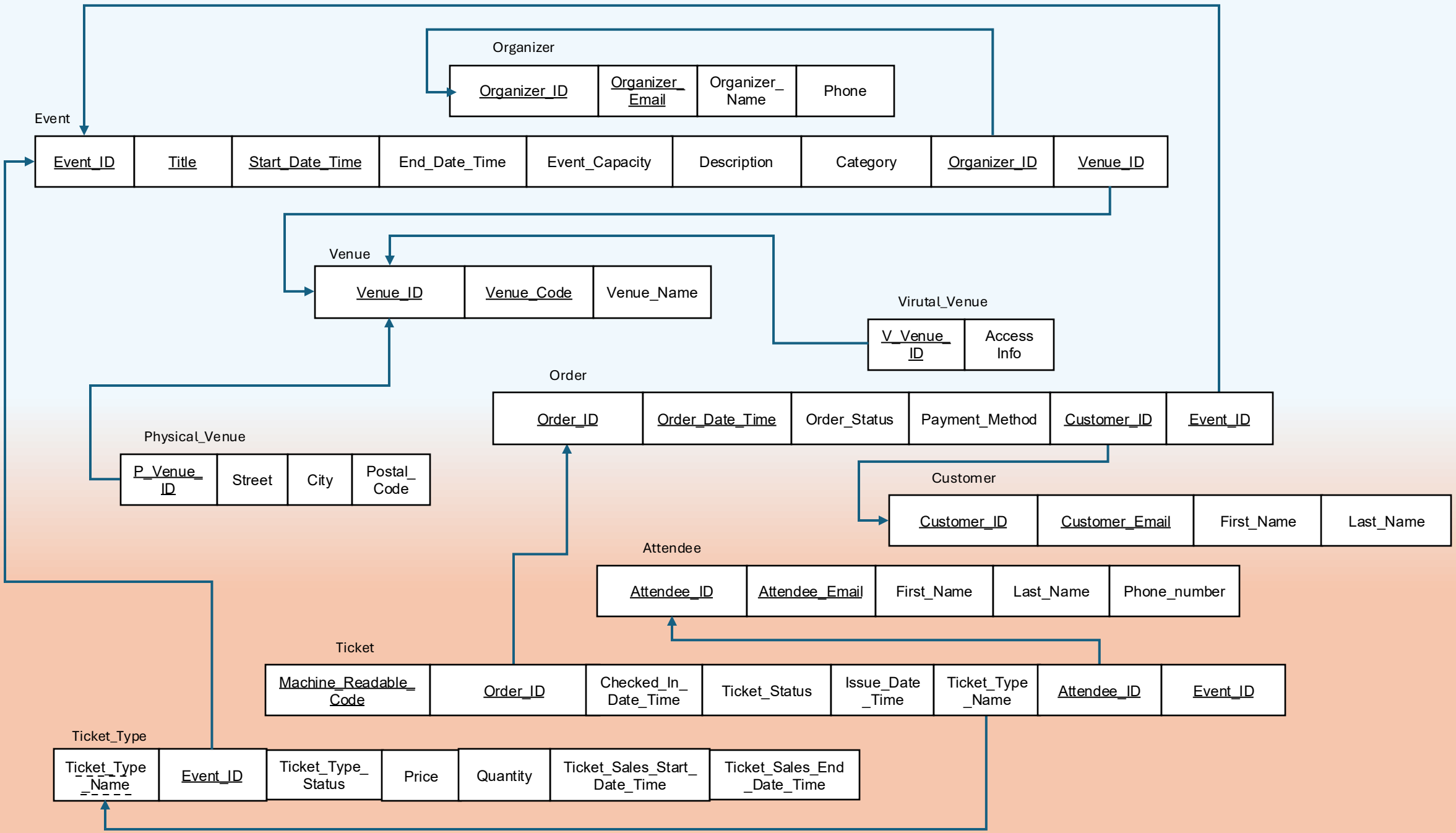


Weak Entities





Logical Model


Organizer
Organizer_ID
Organizer_ Email
Organizer_ Name
Phone
Event
Event_ID
Title
Start_Date_Time
End_Date_Time
Event_Capacity
Description
Category
Organizer_ID
Venue_ID
Venue
Venue_ID
Venue_Code
Venue_Name
Virutal_Venue
V_Venue_ ID
Access Info
Order
Order_ID
Order_Date_Time
Order_Status
Payment_Method
Customer_ID
Event_ID
Physical_Venue
P_Venue_ ID
Street
City
Postal_ Code
Customer
Customer_ID
Customer_Email
First_Name
Last_Name
Attendee
Attendee_ID
Attendee_Email
First_Name
Last_Name
Phone_number
Ticket
Machine_Readable_ Code
Order_ID
Checked_In_ Date_Time
Ticket_Status
Issue_Date _Time
Ticket_Type _Name
Attendee_ID
Event_ID
Ticket_Type
Ticket_Type _Name
Event_ID
Ticket_Type_ Status
Price
Quantity
Ticket_Sales_Start_ Date_Time
Ticket_Sales_End _Date_Time

Database Design & Implementation

Event

EventID (PK)	Title	StartDateTime	EndDateTime	EventCapacity	Description	Category	OrganizerID (FK)	VenueID (FK)
1	Innovation Summit	2026-04-12 09:00:00	2026-04-12 17:00:00	450	Technology and innovation speakers and workshops	Technology	1	1
2	Summer Sounds Live	2026-06-18 15:00:00	2026-06-18 22:30:00	1800	Live outdoor music event	Music	2	4
3	Leadership Workshop	2026-05-08 08:30:00	2026-05-08 15:30:00	220	Professional leadership development sessions	Business	1	2
4	Local Makers Market	2026-07-09 10:00:00	2026-07-09 18:00:00	300	Showcase of local creators and small businesses	Community	3	3
5	Design & Creativity Expo	2026-09-14 10:00:00	2026-09-14 16:00:00	275	Panels and exhibits focused on creative industries	Art	3	1
6	Data Analytics Bootcamp	2026-10-03 09:00:00	2026-10-03 16:00:00	180	One-day bootcamp covering analytics fundamentals and dashboards	Education	2	2

Organizer

OrganizerID (PK)	OrganizerEmail	OrganizerName	Phone
1	info@northstarevents.com	NorthStar Events	801-555-1101
2	contact@summitproductions.com	Summit Productions	801-555-1102
3	hello@communitylive.org	Community Live	801-555-1103

Venue

VenueID (PK)	VenueCode	VenueName
1	CNV-01	Central Convention Center
2	THR-02	Grand Theater
3	HALL-03	Riverside Hall
4	OUT-04	Lakeside Amphitheater

Event Order

OrderID (PK)	OrderDateTime	OrderStatus	PaymentMethod	CustomerID (FK)	EventID (FK)
1001	2026-03-01 09:10:00	Completed	Credit Card	1	1
1002	2026-03-02 11:25:00	Completed	PayPal	2	1
1003	2026-03-03 14:40:00	Completed	Credit Card	3	2
1004	2026-03-04 16:05:00	Pending	Debit Card	4	3
1005	2026-03-05 10:20:00	Completed	Credit Card	5	4
1006	2026-03-06 13:55:00	Completed	PayPal	6	5
1007	2026-03-07 09:45:00	Completed	Credit Card	1	2
1008	2026-03-08 12:15:00	Completed	Credit Card	2	6

Customer

CustomerID (PK)	CustomerEmail	FirstName	LastName
1	j.adams@email.com	Jordan	Adams
2	m.turner@email.com	Morgan	Turner
3	c.hall@email.com	Casey	Hall
4	t.wright@email.com	Taylor	Wright
5	a.reed@email.com	Alex	Reed
6	k.evans@email.com	Kris	Evans

Attendee

AttendeeID (PK)	AttendeeEmail	FirstName	LastName	PhoneNumber
1	jordana@email.com	Jordan	Adams	801-555-2201
2	emilyb@email.com	Emily	Brooks	801-555-2202
3	samk@email.com	Sam	Knight	801-555-2203
4	oliviar@email.com	Olivia	Ross	801-555-2204
5	liamp@email.com	Liam	Parker	801-555-2205
6	norahm@email.com	Norah	Mitchell	801-555-2206
7	ethanj@email.com	Ethan	James	801-555-2207
8	sophiat@email.com	Sophia	Taylor	801-555-2208
9	lucasw@email.com	Lucas	Ward	801-555-2209
10	gracef@email.com	Grace	Foster	801-555-2210
11	m.turner@email.com	Morgan	Turner	801-555-2211

Ticket

MachineReadable Code (PK)	OrderID (FK)	CheckedIn DateTime	TicketStatus	IssueDateTime	TicketTypeName (FK)	AttendeeID (FK)	EventID (FK)
T-1001-A	1001		Issued	2026-03-01 09:11:00	General Admission	1	1
T-1001-B	1001		Issued	2026-03-01 09:11:00	VIP Access	2	1
T-1002-A	1002		Issued	2026-03-02 11:26:00	General Admission	3	1
T-1003-A	1003		Issued	2026-03-03 14:41:00	General Admission	4	2
T-1003-B	1003		Issued	2026-03-03 14:41:00	Front Section	5	2
T-1004-A	1004		Issued	2026-03-04 16:06:00	Standard Pass	6	3
T-1005-A	1005		Issued	2026-03-05 10:21:00	Entry Pass	7	4
T-1006-A	1006		Issued	2026-03-06 13:56:00	Expo Pass	8	5
T-1007-A	1007		Issued	2026-03-07 09:46:00	General Admission	9	2
T-1007-B	1007		Issued	2026-03-07 09:46:00	Front Section	10	2
T-1008-A	1008		Issued	2026-03-08 12:16:00	VIP Access	11	6
T-1008-B	1008		Issued	2026-03-08 12:16:00	VIP Access	4	6

Ticket Type

EventID (PK)	TicketTypeName (PK)	TicketTypeStatus	Price	Quantity	TicketSales StartDateTime	TicketSales EndDateTime
1	General Admission	Active	65.0000	350	2026-01-0100:00:00	2026-04-11 23:59:59
1	VIP Access	Active	150.0000	100	2026-01-01 00:00:00	2026-04-11 23:59:59
2	Front Section	Active	160.0000	300	2026-02-01 00:00:00	2026-06-17 23:59:59
2	General Admission	Active	85.0000	1400	2026-02-01 00:00:00	2026-06-17 23:59:59
3	Standard Pass	Active	45.0000	220	2026-02-15 00:00:00	2026-05-07 23:59:59
4	Entry Pass	Active	20.0000	300	2026-03-01 00:00:00	2026-07-08 23:59:59
5	Expo Pass	Active	30.0000	275	2026-04-01 00:00:00	2026-09-13 23:59:59
6	Standard Pass	Active	55.00	140	2026-07-01 00:00:00	2026-10-02 23:59:59
6	VIP Access	Active	95.00	40	2026-07-01 00:00:00	2026-10-02 23:59:59

Database Demonstration

Create a number of SQL statements (inserts, updates, selects, deletes) to demonstrate each “Completed” requirement that is supported by the database

- Add a new event for that organizer
- Add a new customer and order
- Add a new event for that organizer
- Update an event’s capacity
- Change an order status from Pending → Completed
- Update ticket status when attendee checks in
- List all events with organizer and venue names
- Count tickets sold per event
- Find all attendees who bought VIP Access for Innovation Summit
- Deletes row, Delete a ticket

Add a new organizer

```
INSERT INTO Event101.Organizer (OrganizerID, OrganizerEmail, OrganizerName, Phone)
VALUES (10, 'events@utahcreatives.org', 'Utah Creatives', '801-555-9911');
```

OrganizerID	OrganizerEmail	OrganizerName	Phone
1	info@northstarevents.c...	NorthStar Events	801-555-1101
2	contact@summitprodu...	Summit Productions	801-555-1102
3	hello@communitylive.o...	Community Live	801-555-1103
10	events@utahcreatives....	Utah Creatives	801-555-9911

Add a new event for that organizer

```
INSERT INTO Event101.Event (EventID, Title, StartDateTime, EndDateTime, EventCapacity,
 Description, Category, OrganizerID, VenueID)
VALUES (10, 'Art in the Park', '2026-10-01 10:00:00', '2026-10-01 17:00:00',500, 'Outdoor art festival', 'Art',
10,  4);
```

EventID	Title	StartDateTime
10	Art in the Park	2026-10-01T10:00:00.0000000

Add a new customer and order

```
INSERT INTO Event101.EventOrder (OrderID, OrderDateTime, OrderStatus, PaymentMethod,
 CustomerID, EventID)
VALUES (1234, '2026-01-10 12:00:00', 'Completed', 'Credit Card', 39, 3);

INSERT INTO Event101.Customer (CustomerID, CustomerEmail, FirstName, LastName)
VALUES (39, 'p.lopez@email.com', ‘Gat', 'Lopez');
```

SELECT* FROM EVENT101.Organizer

OrderID	OrderDateTime	OrderStatus	PaymentMet
1234	2026-01-10T12:00:0...	Completed	Credit Card

SELECT* FROM EVENT101.Organizer

CustomerID	CustomerEmail	FirstName	LastName
39	p.lopez@email.com	Gat	Lopez

Add a ticket type and ticket

```
INSERT INTO Event101.TicketType (EventID, TicketTypeName, TicketTypeStatus, Price,
Quantity, TicketSalesStartDateTime, TicketSalesEndDateTime)
VALUES (10, 'General Admission', 'Active', 25.00, 500,
 '2026-05-01 00:00:00', '2026-09-30 23:59:59');

INSERT INTO Event101.Ticket (MachineReadableCode, OrderID, CheckedInDateTime,
 TicketStatus, IssueDateTime, TicketTypeName, AttendeeID, EventID)
VALUES ('T-2001-A', 2001, NULL, 'Issued', '2026-03-10 12:01:00',
 'General Admission', 1, 10);
```

SELECT* FROM EVENT101.Organizer

EventID	TicketTypeName	TicketTypeStatus	Price	Quantity
10	General Admission	Active	25.0000	500

MachineReadableC...	OrderID	CheckedInDateTime	TicketStatus	IssueD
T-2001-A	2001		Issued	2026-

Update an event's capacity

```
UPDATE Event101.Event
SET EventCapacity = 600
WHERE EventID = 10;
```

SELECT* FROM EVENT101.Organizer

EventID	Title	StartDateTime	EndDateTime	EventCapacity
10	Art in the Park	2026-10-01T10:00:00.0000000	2026-10-01T17:00:00.0000000	600

Change an order status from Pending → Completed

```
UPDATE Event101.EventOrder
SET OrderStatus = 'Completed'
WHERE OrderID = 1004;
```

OrderID	OrderDateTime	OrderStatus	PaymentMethod	CustomerID	EventID
1004	2026-03-04T16:05:00.0...	Completed	Debit Card	4	3

Update ticket status when attendee checks in

```
UPDATE Event101.Ticket
SET TicketStatus = 'Checked In',
   CheckedInDateTime = GETDATE()
WHERE MachineReadableCode = 'T-1003-A';
```

MachineReadableC...	OrderID	CheckedInDateTime	TicketStatus	IssueDateTime	TicketTypeName	AttendeeID
T-1003-A	1003	2026-02-28T02:00:5...	Checked In	2026-03-03T14:41:0...	General Admission	4

List all events with organizer and venue names

```
SELECT e.EventID, e.Title, e.StartDateTime, e.EndDateTime,
     o.OrganizerName, v.VenueName
FROM Event101.Event e
JOIN Event101.Organizer o ON e.OrganizerID = o.OrganizerID
JOIN Event101.Venue v ON e.VenueID = v.VenueID;
```

SELECT* FROM EVENT101.Organizer

EventID	Title	StartDateTime	EndDateTime	OrganizerName	VenueName
1	Innovation Summit	2026-04-12T09:00:00.0...	2026-04-12T17:00:00.0...	NorthStar Events	Central Convention Cei
2	Summer Sounds Live	2026-06-18T15:00:00.0...	2026-06-18T22:30:00.0...	Summit Productions	Lakeside Amphitheater
3	Leadership Workshop	2026-05-08T08:30:00.0...	2026-05-08T15:30:00.0...	NorthStar Events	Grand Theater

Count tickets sold per event

```
SELECT EventID, COUNT(*) AS TicketsSold
FROM Event101.Ticket
GROUP BY EventID;
```

SELECT* FROM EVENT101.Organizer

EventID	TicketsSold
1	3
2	4
3	1

Find all attendees who bought VIP Access for Innovation Summit

```
SELECT a.AttendeeID, a.FirstName, a.LastName
FROM Event101.Ticket t
JOIN Event101.Attendee a ON t.AttendeeID = a.AttendeeID
WHERE t.EventID = 1
  AND t.TicketTypeName = 'VIP Access';
```

AttendeeID	FirstName	LastName
2	Emily	Brooks

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